

Overview of Time series forecasting

Subject: Time series forecasting

1.

A simple way to examine a regular time series is manually with a line chart. The datagraphic shows tuberculosis deaths in the United States, along with the yearly change and the percentage change from year to year. The total number of deaths declined in every year until the mid-1980s, after which there were occasional increases, often proportionately - but not absolutely - quite large. A study of corporate data analysts found two challenges to exploratory time series analysis: discovering the shape of interesting patterns, and finding an explanation for these patterns. Visual tools that represent time series data as heat map matrices can help overcome these challenges.

2.

Curve fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points, possibly subject to constraints. Curve fitting can involve either interpolation, where an exact fit to the data is required, or smoothing, in which a "smooth" function is constructed that approximately fits the data. A related topic is regression analysis, which focuses more on questions of statistical inference such as how much uncertainty is present in a curve that is fit to data observed with random errors. Fitted curves can be used as an aid for data visualization, to infer values of a function where no data are available, and to summarize the relationships among two or more variables. Extrapolation refers to the use of a fitted curve beyond the range of the observed data, and is subject to a degree of uncertainty since it may reflect the method used to construct the curve as much as it reflects the observed data.

3.

History Jevons was the first to study times series. In his book he examined a number of weekly financial time series between the years 1825 to 1860 inclusive. These included times series of bankruptcies, currency circulation and discount rates.

4.

Consideration of the autocorrelation function and the spectral density function (also cross-correlation functions and cross-spectral density functions) Scaled cross- and auto-correlation functions to remove contributions of slow components Performing a Fourier transform to investigate the series in the frequency domain Performing a clustering analysis Discrete, continuous or mixed spectra of time series, depending on whether the time series contains a (generalized) harmonic signal or not Use of a filter to remove unwanted noise Principal component analysis (or empirical orthogonal function analysis) Singular spectrum analysis "Structural" models: General state space models Unobserved components models Machine learning Artificial neural networks Support vector machine Fuzzy logic Gaussian process Genetic programming Gene expression programming Hidden Markov model Multi expression programming Queueing theory analysis Control chart Shewhart individuals control chart CUSUM chart EWMA chart Detrended fluctuation analysis Nonlinear mixed-effects modeling Dynamic time warping Dynamic Bayesian network Time-frequency analysis techniques: Fast Fourier transform Continuous wavelet transform Short-time Fourier transform Chirplet transform Fractional

Fourier transform Chaotic analysis Correlation dimension Recurrence plots Recurrence quantification analysis Lyapunov exponents Entropy encoding

5.

In general, a function approximation problem asks us to select a function among a well-defined class that closely matches ("approximates") a target function in a task-specific way. One can distinguish two major classes of function approximation problems: First, for known target functions, approximation theory is the branch of numerical analysis that investigates how certain known functions (for example, special functions) can be approximated by a specific class of functions (for example, polynomials or rational functions) that often have desirable properties (inexpensive computation, continuity, integral and limit values, etc.). Second, the target function, call it g , may be unknown; instead of an explicit formula, only a set of points (a time series) of the form $(x, g(x))$ is provided. Depending on the structure of the domain and codomain of g , several techniques for approximating g may be applicable. For example, if g is an operation on the real numbers, techniques of interpolation, extrapolation, regression analysis, and curve fitting can be used. If the codomain (range or target set) of g is a finite set, one is dealing with a classification problem instead. A related problem of online time series approximation is to summarize the data in one-pass and construct an approximate representation that can support a variety of time series queries with bounds on worst-case error. To some extent, the different problems (regression, classification, fitness approximation) have received a unified treatment in statistical learning theory, where they are viewed as supervised learning problems.

6.

Subsequence time series clustering Subsequence time series clustering resulted in unstable (random) clusters induced by the feature extraction using chunking with sliding windows. It was found that the cluster centers (the average of the time series in a cluster - also a time series) follow an arbitrarily shifted sine pattern (regardless of the dataset, even on realizations of a random walk). This means that the found cluster centers are non-descriptive for the dataset because the cluster centers are always nonrepresentative sine waves.

7.

Motivation In the context of statistics, econometrics, quantitative finance, seismology, meteorology, and geophysics the primary goal of time series analysis is forecasting. In the context of signal processing, control engineering and communication engineering it is used for signal detection. Other applications are in data mining, pattern recognition and machine learning, where time series analysis can be used for clustering, classification, query by content, anomaly detection as well as forecasting.

8.

Panel data A time series is one type of panel data. Panel data is the general class, a multidimensional data set, whereas a time series data set is a one-dimensional panel (as is a cross-sectional dataset). A data set may exhibit characteristics of both panel data and time series data. One way to tell is to ask what makes one data record unique from the other records. If the answer is the time data field, then this is a time series data set candidate. If determining a unique record requires a time data field and an additional identifier which is unrelated to time (e.g. student ID, stock symbol, country code), then it is panel data candidate. If the differentiation lies on the non-time identifier, then the data set is a cross-sectional data set candidate.